Case Study Gas Station Sales Analysis & Forecasting

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Case Study

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Introduction

In this case study, I performed a data analysis project for a gas station convenience store to understand sales trends and identify opportunities for revenue growth. The store manager recently took over the business and wanted insights into historical sales performance as well as guidance on improving future sales.

To answer these questions, I followed the **six phases of the data analysis process: Ask, Prepare, Process, Analyze, Share, and Act**. This structured approach helped transform raw transaction data into meaningful business insights and recommendations.

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1. Ask Phase

Business Task

The store manager wanted to better understand how the store has performed over the past several years and how data could help improve future sales.

The objectives of this analysis were to:

- Identify **sales trends over time**
- Determine **which departments generate the most revenue**
- Understand **monthly sales patterns**
- Identify **low-performing products**
- Build a **model to predict future sales**
- Provide **recommendations to increase revenue**

Stakeholders

Store Manager

- Needs insights to improve business performance
- Wants to know which products to prioritize

Data Analyst (Project Role)

- Analyze historical sales data
- Identify patterns and trends
- Provide actionable business recommendations

Key Business Questions

1. During which months does the store experience the highest sales activity?
2. Which department generates the most revenue?
3. Which departments generate the least revenue?
4. What strategies could improve sales during slower months?
5. Can past sales data help predict future sales?

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2. Prepare Phase

Data Source

The dataset used in this analysis was exported from the store's **Elister point-of-sale system**, which records all transactions made at the gas station.

The dataset includes approximately **six years of sales data (2020–2025)**.

Key data fields include:

- Transaction date
- Product category
- Item description
- Sales amount

Data Organization

The data was organized into yearly datasets and summarized using pivot tables to analyze:

- Total sales by year
- Monthly sales trends
- Revenue by department

Data Credibility

The dataset is considered reliable because:

- It originates directly from the store’s POS system
- Transactions are automatically recorded
- The data reflects real customer purchases

However, the dataset has some limitations:

- Profit margins are not included
- Customer demographic information is unavailable
- External factors such as fuel price fluctuations are not included

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3. Process Phase

Tools Used

The following tools were used to clean and analyze the data:

Microsoft Excel

- Data cleaning
- Pivot tables
- Data visualization

Python

- Machine learning model for sales prediction

Data Cleaning

Before analysis, the following data cleaning steps were performed:

- Removed duplicate records
- Checked for missing values
- Removed unnecessary columns
- Standardized product category names
- Verified numeric data formats

Data Transformation

The cleaned data was organized into structured formats to support analysis, including:

- Yearly revenue summaries
- Monthly sales tables
- Department revenue comparisons

Pivot tables were used to aggregate and summarize sales data.

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4. Analyze Phase

Sales Trends Over Time

Year	Total Sales
2020	$127,684
2021	$238,108
2022	$425,090
2023	$403,354
2024	$377,026
2025	$409,205

Key Insight

Sales increased significantly from **2020 to 2022**, suggesting strong business growth during this period.

Sales slightly declined in **2023 and 2024**, but recovered in **2025**, indicating stabilization in revenue.

Monthly Sales Patterns

Monthly sales charts reveal seasonal trends in customer activity.

Higher sales occur during:

- **Spring months (March–May)**
- **Fall months (September–October)**

Lower sales typically occur during:

- **Winter months (January–February)**

These patterns may be influenced by travel behavior and seasonal demand.

Department Revenue Analysis

The departments contributing the most revenue are:

Department	Total Revenue
Regular Gas	$10,166,304
Cigarettes	$981,179
Lottery Sales	$829,651
Beer	$522,838
Tobacco	$486,436

Key Insight

Regular gas generates the largest share of revenue, which is typical for gas stations.

However, convenience store products such as cigarettes, beer, and lottery tickets also generate substantial revenue.

Lowest Performing Department

Department	Revenue
Cigarette Carton	$4,202

Cigarette cartons generated the lowest revenue, indicating low customer demand for this product category.

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5. Share Phase

To communicate insights clearly, several visualizations were created:

- Monthly sales trend charts for each year
- Year-over-year sales comparison
- Department revenue distribution
- Pivot table summaries

These visualizations help stakeholders quickly identify patterns and understand business performance.

Visualizations Images:

1. **Pivot Tables**

2020 Sales

Row Labels	Sum of Amount
Air Vac	1144
Autoparts	11.98
Beer	101123.25
Cigarette	255693.68
Cigarette Carton	1745.41
Fountain/Coffee	3898.77
Lotto Sales	133737
Philip Morris USA Inc	8748.18
Sales Non-Tax	10206.51
Sales Tax	34080.66
Sales Taxable	124136.94
Scratchoff Sales	601999
US Smokeless Tobacco Brand	58.95
Grand Total	**1276584.33**

2021 Sales

Row Labels	Sum of Amount
Air Vac	1575
Autoparts	32.33
Beer	104952.5
Cigarette	287047.48
Cigarette Carton	30.55
Fountain/Coffee	166.05
Lotto Sales	171925.5
MID	68448.09
PRE	109515.29
REG	729051.8
Sales Non-Tax	10233.42
Sales Tax	41803.86
Sales Taxable	142234.11
Scratchoff Sales	651645
Soda	4572.58
Tobacco	57796.14
US Smokeless Tobacco Brand	56.5
Grand Total	**2381086.2**

2022 Sales

Row Labels	Sum of Amount
Air Vac	1593
Autoparts	1128.53
Beer	123764.58
Cigarette	204967.33
Cigarette Carton	4032.49
Fountain/Coffee	872.22
Lotto Sales	184428.5
PRE	351917.26
REG	2465882.01
Sales Non-Tax	12344.76
Sales Tax	45601.05
Sales Taxable	140935.47
Scratchoff Sales	539396
Soda	31972.72
Tobacco	142124.75
Grand Total	**4250960.67**

2023 Sales

Row Labels	Sum of Amount
Air Vac	2520
Autoparts	2268.29
Beer	104105.93
Cigarette	188473.58
Cigarette Carton	37.44
Fountain/Coffee	1006.19
Lotto Sales	171067.5
PRE	318297.33
Pump Test	-11.7
REG	2390012.83
Sales Non-Tax	15125.88
Sales Tax	41909.72
Sales Taxable	97254.92
Scratchoff Sales	499916
Soda	71999.5
Tobacco	129562.02
Grand Total	**4033545.43**

2024 Sales

Row Labels	Sum of Amount
Air Vac	1213
Autoparts	3020.65
Beer	92804.45
Cigarette	156914.71
Fountain/Coffee	1882.6
MID	214907.31
REG	2481562.14
Sales Non-Tax	20597.03
Sales Tax	36932.55
Sales Taxable	93588.03
Scratchoff Sales	489605
Soda	86666.97
Tobacco	90561.51
Grand Total	**3770255.95**

2025 Sales

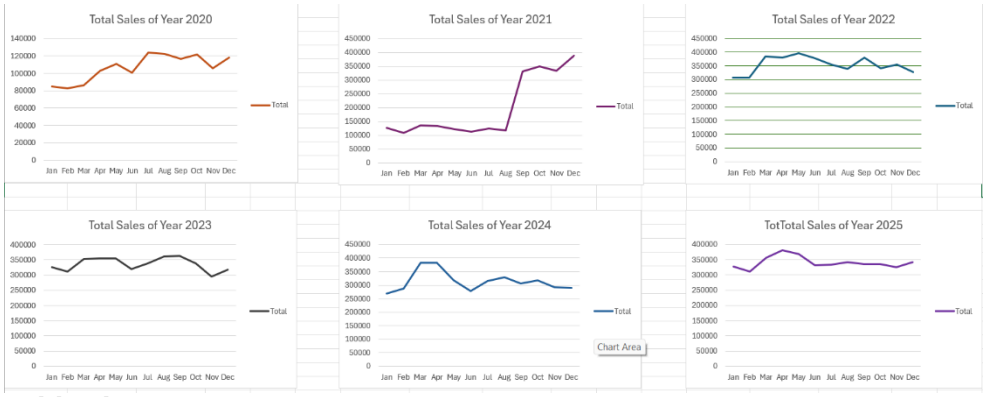
Row Labels	Sum of Amount
Air Vac	201
Autoparts	2450.72
Beer	97211.4
Cigarette	143776.13
Cigarette Carton	101.97
Fountain/Coffee	7958.73
Lotto Sales	302229.95
MID	206288.94
PRE	333629.39
REG	2099795.39
Sales Non-Tax	22460.19
Sales Tax	39245.69
Sales Taxable	154191.23
Scratchoff Sales	534207
Soda	81913.49
Tobacco	66392.03
Grand Total	**4092053.25**

2. **Total revenue by Department**

Total Revenue Generated By Department (2020-2025)

Department	Total revenue by Department
Air Vac	7102
Autoparts	8900.52
Beer	522838.86
Cigarette	981179.23
Cigarette Carton	4202.45
Fountain/Coffee	11885.79
Lotto Sales	829651.45
MID	489644.34
PRE	1113359.27
REG	10166304.17
Sales Non-Tax	80761.28
Sales Tax	205492.87
Sales Taxable	628203.76
Scratchoff Sales	2714769
Soda	277125.26
Tobacco	486436.45
Most Revenue Generated Amount:	10166304.17
Least Revenue Generated Amount	4202.45

3. **Line charts**



4. **Model Pridiction Image:**

```
# first few rows of the future predictions
print(future_df.head())

        Date Item_Category Item_Description  year  month  day  day_of_week
0 2026-01-01      DeptSale    Sales Taxable  2026      1    1            3
1 2026-01-01      DeptSale    Sales Non-Tax  2026      1    1            3
2 2026-01-01      DeptSale  Fountain/Coffee  2026      1    1            3
3 2026-01-01      DeptSale        Cigarette  2026      1    1            3
4 2026-01-01      DeptSale             Beer  2026      1    1            3

   Predicted_Sales
0        206.66910
1         17.57450
2          4.61150
3        492.85470
4        182.14815
```

```
#Aggregate daily sales predictions
daily_sales = future_df.groupby('Date')['Predicted_Sales'].sum()

print(daily_sales.head())

Date
2026-01-01    11570.23945
2026-01-02    13947.30745
2026-01-03    13542.05980
2026-01-04    10557.02410
2026-01-05    12084.84160
Name: Predicted_Sales, dtype: float64
```

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6. Act Phase

Based on the analysis, several strategies could help improve the store's sales performance.

Recommendation 1: Promote High-Margin Store Products

Convenience store products typically have higher profit margins than fuel.

The store should promote items such as:

- Drinks
- Snacks
- Coffee
- Beer

Example promotion:

Fuel purchase + discounted drink

This can encourage customers to enter the store after fueling.

Recommendation 2: Increase Sales During Slow Months

Sales are lowest during winter months. Promotional strategies could include:

- Coffee promotions
- Bundle deals
- Seasonal discounts

Example promotion:

Coffee + snack combo deals

Recommendation 3: Optimize Product Inventory

Low-performing products such as cigarette cartons should be reviewed.

Possible actions include:

- Replacing them with higher-demand items
- Offering discounts to clear inventory
- Adjusting stock levels

Conclusion

This analysis examined six years of gas station sales data to uncover trends and identify opportunities for business improvement.

Key findings include:

- Sales grew significantly between 2020 and 2022
- Gas sales dominate revenue, but convenience store items remain important
- Seasonal trends influence monthly sales performance
- Some product categories generate very low sales

Using data-driven strategies such as targeted promotions, improved product placement, and predictive modeling can help the store increase revenue and optimize inventory decisions.

Skills Demonstrated

This project highlights several core data analytics skills:

- Data cleaning and preparation
- Exploratory data analysis
- Pivot tables and aggregation
- Sales trend analysis
- Business insight development
- Machine learning for forecasting

Tools used

- Microsoft Excel
- Python
- Data visualization techniques